



SCUOLA
ALTI STUDI
LUCCA



Open Lab

A gentle introduction to fMRI: basics

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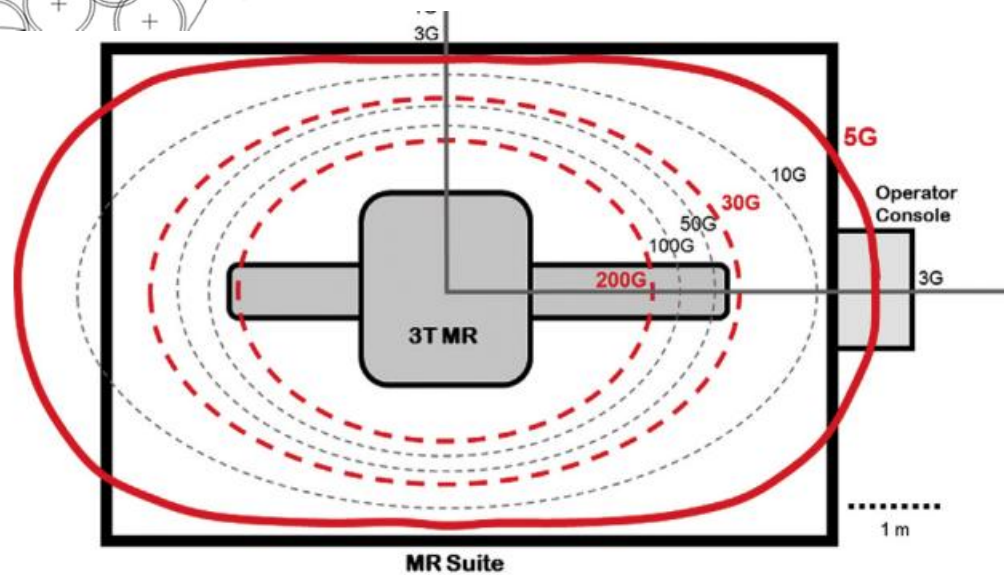




*fMRI is like trying to assemble a ship in a bottle
– every which way you try to move, you
encounter a constraint [Mel Goodale]*

*That's on a good day. On a bad day, it's like
trying to assemble a ship in a bottle
blindfolded, drunk and with one hand tied
behind your back [Jody Culham]*

- Part 1: Univariate Analysis **fMRI basics**
 - fMRI experimental setup
 - Preprocessing of anatomical datasets
 - Preprocessing of functional datasets
 - OpenNeuro: publicly available datasets
- Part 2: Multivariate Analysis **fMRI advanced**
 - Functional connectivity (graph analysis)
 - Multivariate pattern analysis
 - Encoding/Decoding



unfortunate MRI events



fMRI may NOT be safe or appropriate for people:

- with shrapnel or other metal or electronic implants in their bodies (e.g. pacemakers, aneurysm clips, surgical devices, metallic tattoos on the head, IUD, old dental implants)
- who are (or may be) pregnant
- currently using sleeping aids, painkillers (including aspirin), or other agents known to affect BOLD signals (e.g., antihistamines, decongestants, coffee)
- sickle cell anemia (and thalassemia)
- (with a history of head trauma or fainting)

Before entering the scanner, participants must remove all metal items from the body (e.g., glasses, hair clips, jewelry, piercings, makeup, watches, belts) and remove any clothing (including bras) containing metal (snaps, zippers, buttons, hooks, wires, etc.)

Resonance Regional Medical Center Magnetic Resonance Imaging (MRI) Patient Informed Consent

Your physician has ordered an MRI examination as part of the diagnostic tests for your medical evaluation. This procedure can provide information which will enhance your diagnosis. The procedure is accomplished by using a strong magnetic field, radiofrequency waves and a computer to generate the image. No x-rays or radioactive materials are necessary.

MRI systems have been in use for many years with no harmful side effects reported, except in patients with intracranial metal. Your participation in this examination is voluntary and you may withdraw at any time. Because of the nature of the magnetic field, it is necessary to know if you have had certain surgeries or traumas which could have resulted in any metal implants in the body. Please inform the technologist or nurse if you have any of the following [check the appropriate lines]:

Yes No

- ☐ ☐ 1. Cardiac valve prosthesis
- ☐ ☐ 2. Cardiac pacemaker or pacer wire implants
- ☐ ☐ 3. Intracranial aneurysm or bypass surgery
- ☐ ☐ 4. Middle ear prosthesis
- ☐ ☐ 5. Neuro or bio-stimulator
- ☐ ☐ 6. Joint or limb prosthesis
- ☐ ☐ 7. Old shrapnel or welding wounds or accidents
- ☐ ☐ 8. Other internal metal
- ☐ ☐ 9. Is there any possibility of pregnancy

Qualified medical and technical personnel will be in attendance throughout the entire procedure.

All data collected in this study is confidential as part of your patient record. If the data is used for research studies or publications, no patient identification will appear. This information will be used only for educational, research, and scientific purposes.

Signed _____ Date _____
Patient's Name

Child's Assent _____ Date _____
(if seven or older)

Parent / Guardian

Witness

MRI Informed Consent

IRB Consent Form

[Title of Study]

[Name of Researcher]

[Brief Description of Study's Purpose and Objectives]

I. Introduction

- Explanation of the purpose of the consent form
- Explanation of the nature of the study
- Statement of the voluntary nature of participation
- Explanation that participation in the study is not a requirement
- Explanation of the role of the Institutional Review Board (IRB)
- Explanation of the potential risks and benefits of participating in the study
- Explanation of the privacy and confidentiality protections

II. Procedures

- Explanation of what the participant will be asked to do during the study
- Explanation of any procedures or interventions involved in the study
- Explanation of the anticipated duration of the participant's involvement in the study
- Explanation of any potential risks or discomforts associated with participation in the study
- Explanation of any potential benefits of participation in the study
- Explanation of any alternative treatments or procedures that may be available to the participant.

III. Confidentiality and Privacy

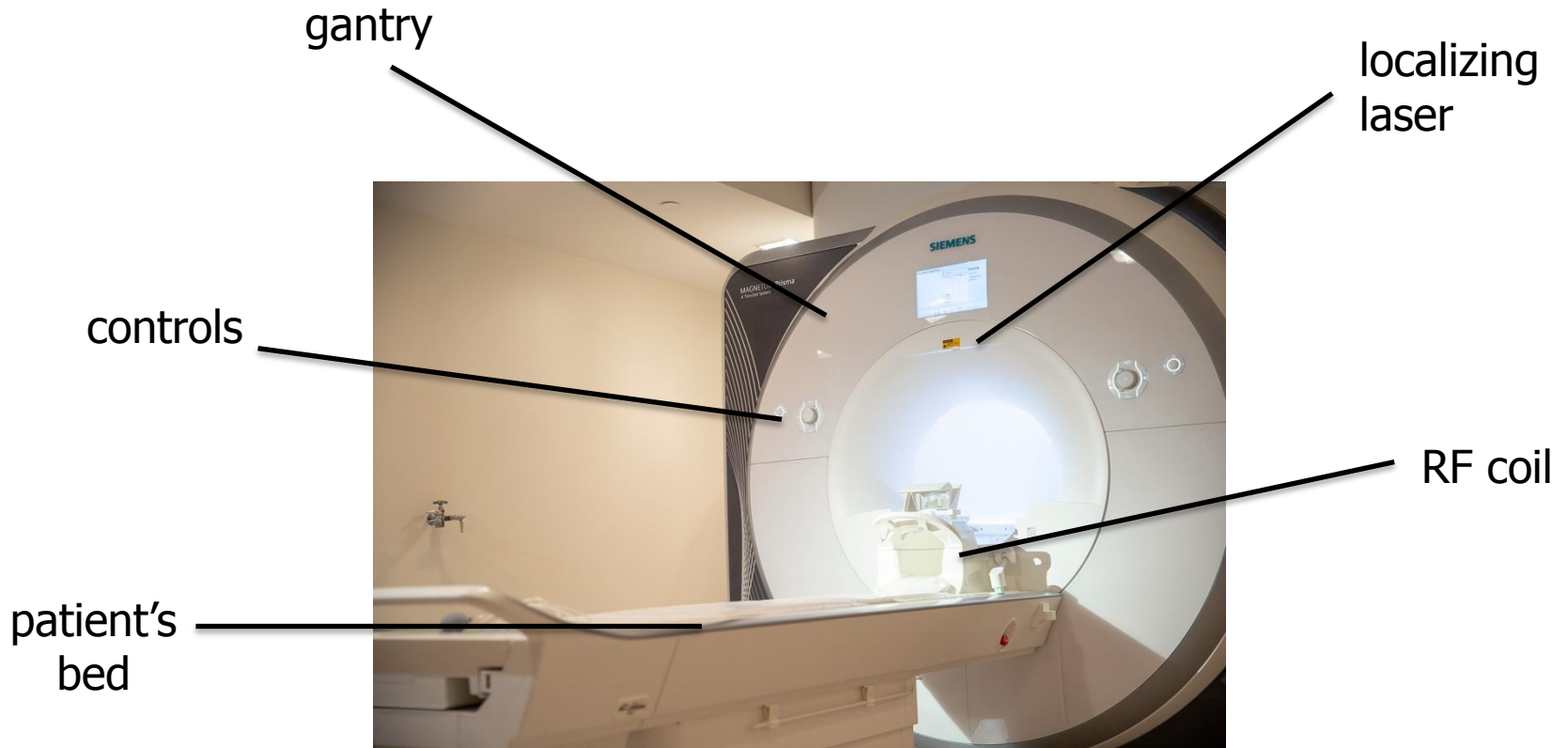
- Explanation of how participant data will be collected, stored, and used during the study
- Explanation of any measures that will be taken to ensure participant confidentiality and privacy
- Explanation of any circumstances under which participant data may be shared with others
- Explanation of any limitations to confidentiality or privacy

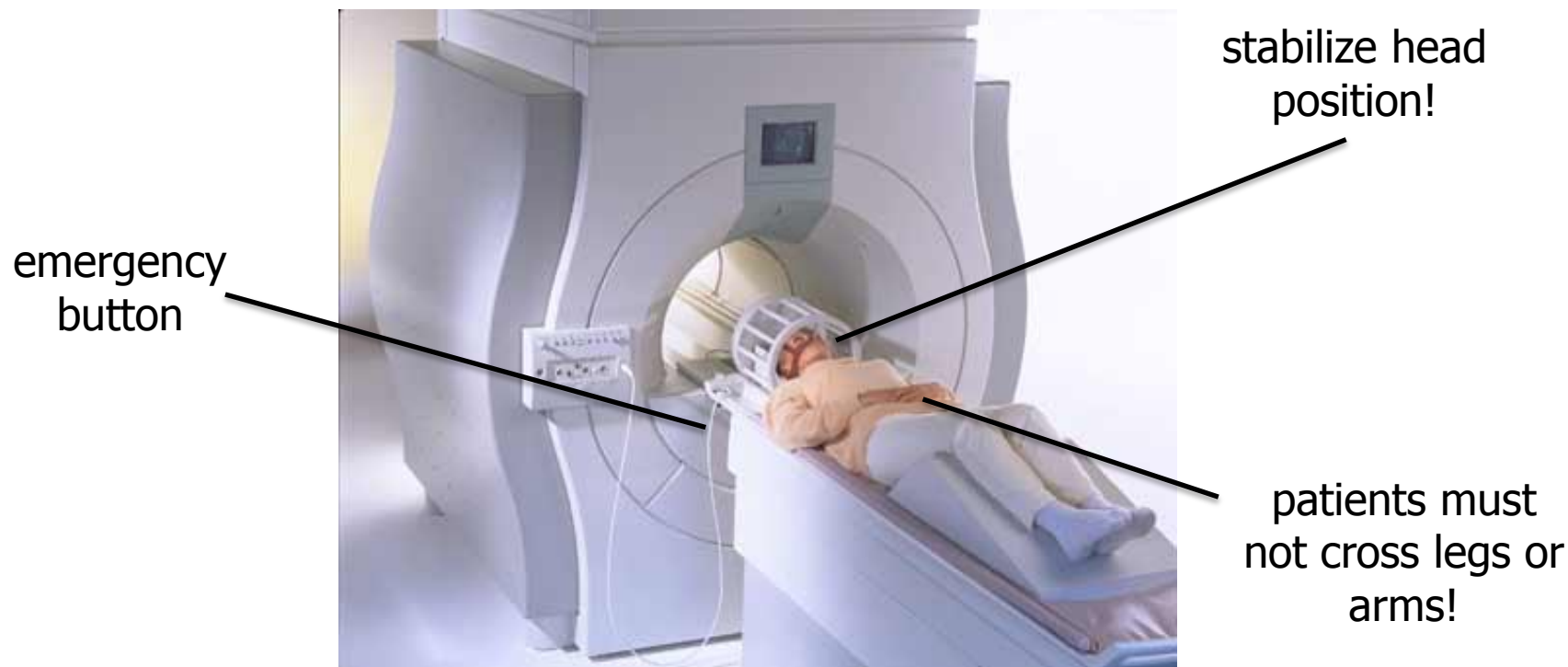
IV. Voluntary Participation and Withdrawal

- Explanation that participation in the study is voluntary and that the participant may withdraw from the study at any time
- Explanation of any potential consequences of withdrawing from the study

V. Benefits and Risks

IRB Informed Consent







foam pads



headcase



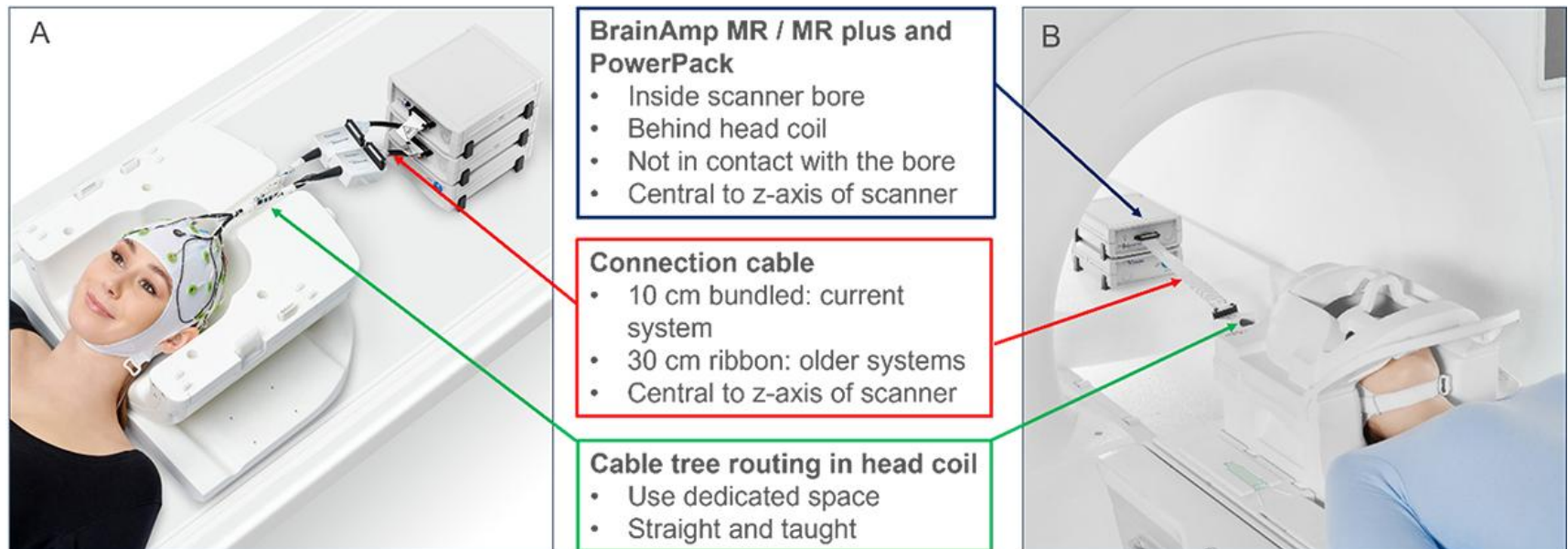
medical tape



pulse oximetry
sensor



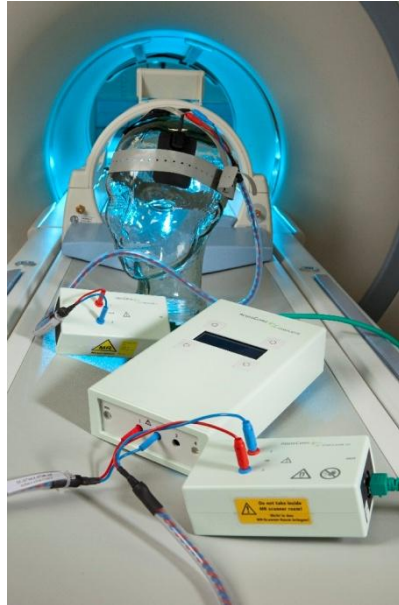
thoracic belt



EEG setup

Cost-benefit analysis:

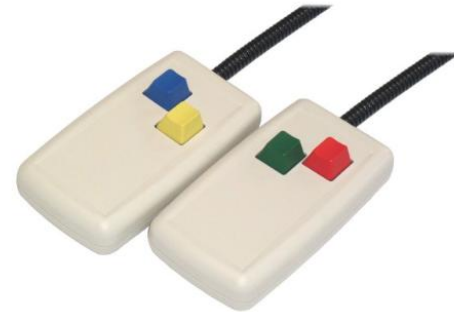
- pros: non-invasive measures of brain activity at varying spatial and temporal scales
- cons: gradient, cardioballistic, scanner (pump/ventilation) artifacts



Neuro stimulation setup

Cost-benefit analysis:

- pros: minimal artefacts on MRI
- cons: tDCS/tACS biological effects (but see Violante et al., 2023)



button box

MRI setup: stimulation



mirror configuration



googles + headphones + microphone



box in scanner room



box in the control room

Cost-benefit analysis:

- pros: minimal artefacts on MRI, diopter correction
- cons: low resolution (800x600 pixels), small visual angle (20°), ghosting and after-effects on the screen



Goggles with eye-tracking

Cost-benefit analysis:

- pros: high resolution (full HD), independent video sources (stereoscopy), eye-tracker
- cons: small visual angle (30°)



Vibro-tactile stimulators

Cost-benefit analysis:

- pros: easy-to-use
- cons: only suitable for hand stimulation



Trigger box

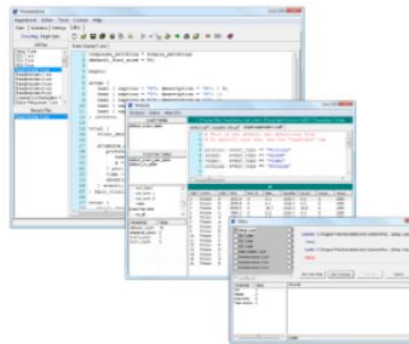


PC to handle button press, visual and
acoustic stimulation



E-Prime

<https://pstnet.com/>



Presentation

<https://www.neurobs.com/>



PsychoPy

<https://www.psychopy.org/>

Box 2.1. fMRI Terminology

Session. The experimental session encompasses the time that the subject enters the scanner until they leave the scanner. This will usually include multiple scanning runs with different pulse sequences, including structural, diffusion imaging, functional MRI, spectroscopy, etc.

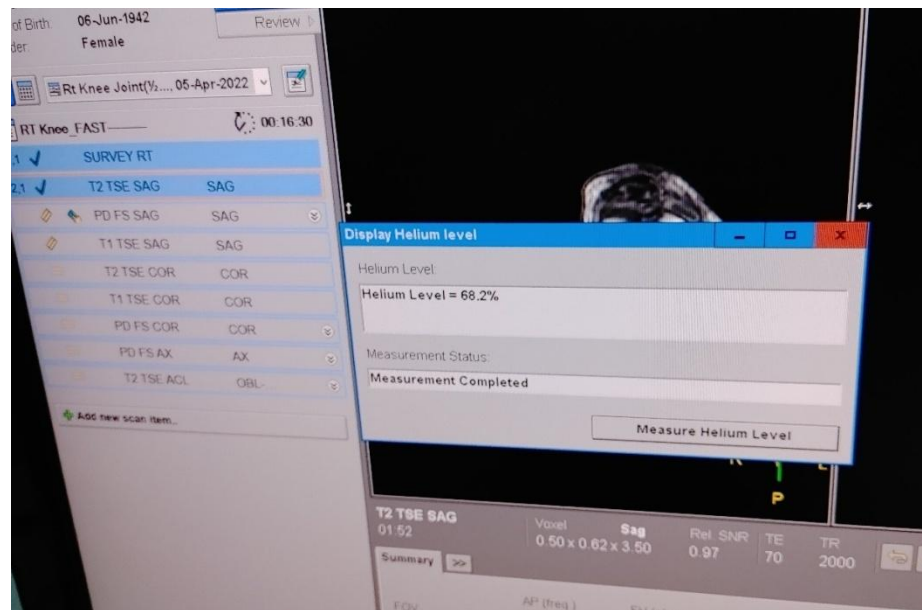
Run. A run is a period of temporally continuous data acquisition using a single pulse sequence.

Volume. A volume (or alternatively “frame”) is single 3-dimensional image acquired as part of a run.

Condition. A condition is a set of task features that are created to engage a particular mental state.

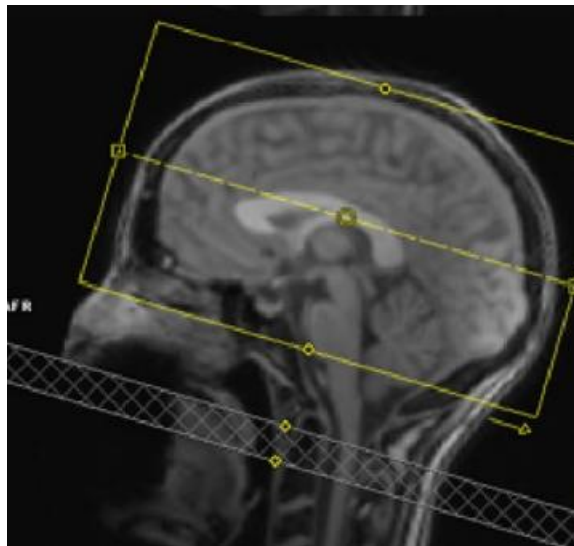
Trial. A trial (or alternatively “event”) is a temporally isolated period during which a particular condition is presented, or a specific behavior is observed.

outside the MRI: scanner software



Summary	Geometry	Contrast	Motion	Dyn/Ang	Postproc	Offc/Ang	Coils	Conflicts	<<
Preparation phases	auto					Total scan duration	05:34.0		
Interactive F0	no					Rel. SNR	1		
MIP/MPR	no					Act. TR (ms)	2000		
Images						Act. TE (ms)	n*13		
Autoview image	no					ACQ matrix M x P	360 x 324		
Calculated images						ACQ voxel MPS (mm)	0.44 / 0.49 / 4.00		
1	T2					REC voxel MPS (mm)	0.29 / 0.29 / 4.00		
2	no					Scan percentage (%)	90.2		
Reference tissue	Skeletal muscle					Packages	1		
Recon. compression	No					Min. slice gap (mm)	0		
Preset window contrast	soft					Act. slice gap (mm)	0.4		
Reconstruction mode	immediate					WFS (pix) / BW (Hz)	1301 / 166.9		
Save raw data	no					TSE es / shot (ms)	13.0 / 104		
Hardcopy protocol	no					Min. TR (ms)	1171		
Image filter	system default					Local torso SAR	< 40 %		
Uniformity correction	no					Whole body SAR / level	< 1.0 W/kg / normal		
Geometry correction	none					SED	< 0.3 kJ/kg		
						Max B1+rms	2.67 uT		
						PNS / level	49 % / normal		
						dB/dt	36.7 T/s		

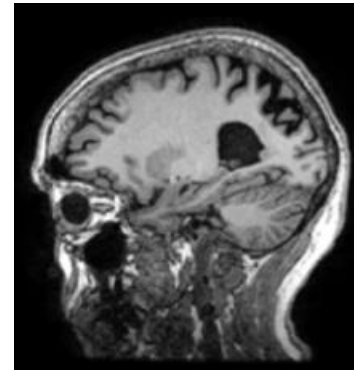
outside the MRI: scanner software



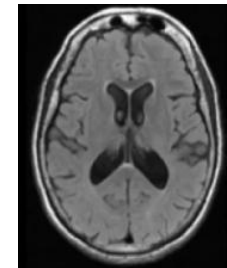
Scout/Survey sequence



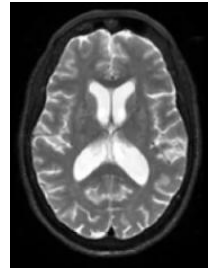
Anatomical
data



T1

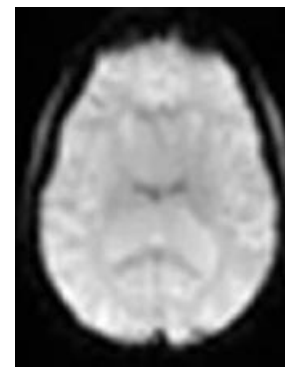


T2-FLAIR



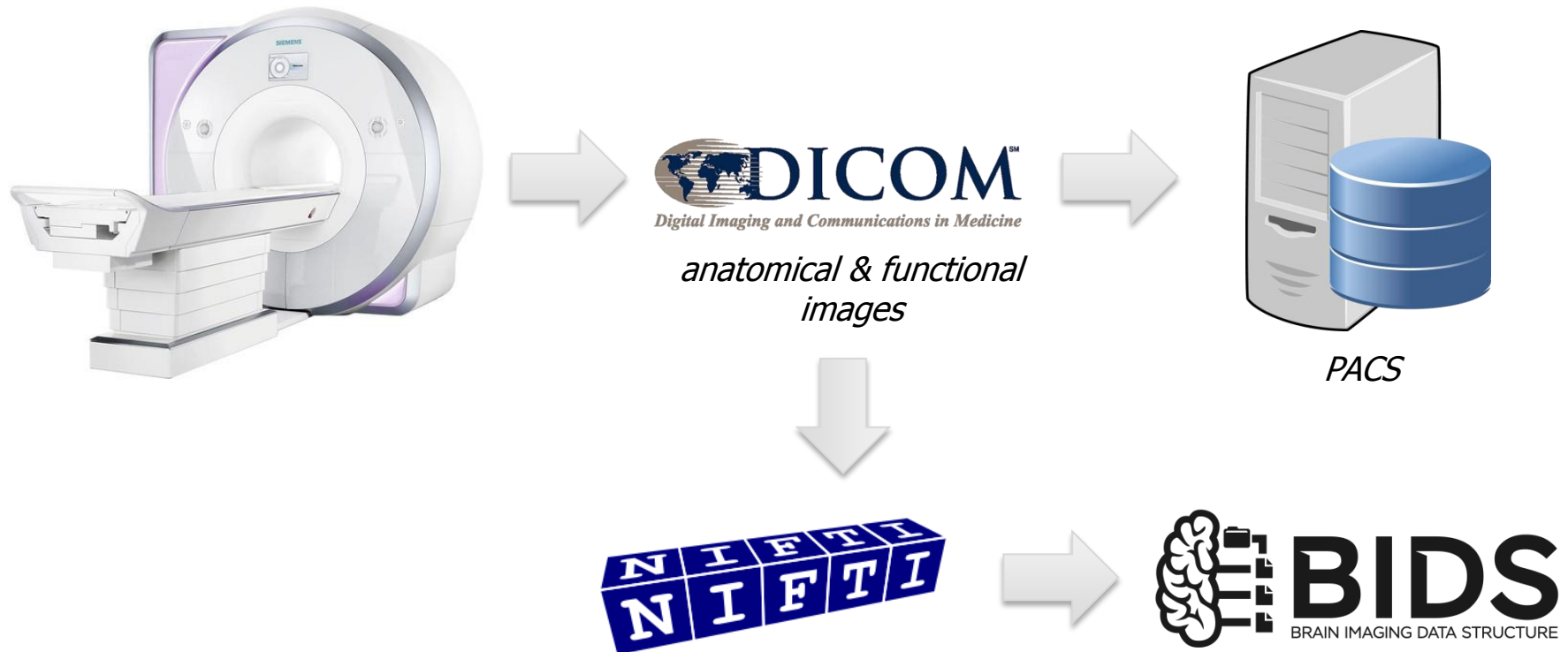
T2

Functional
data



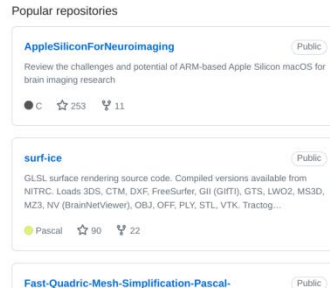
SE/GE EPI
(T2/T2*)

DICOM: Digital Imaging and Communications in Medicine





Chris Rorden
neurolabusc



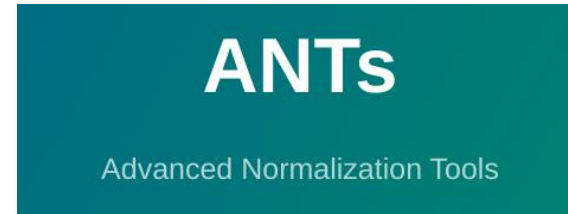
<https://github.com/rordenlab/dcm2niix>



<https://afni.nimh.nih.gov/>



<https://fsl.fmrib.ox.ac.uk/fsl/fslwiki/FSL>



<https://stnava.github.io/ANTs/>



<https://www.fil.ion.ucl.ac.uk/spm/>

```
project/
├── subject
│   ├── session
│   └── datatype
```



```
myProject/
├── sub-01
│   ├── ses-01
│   └── anat
```



	modality						
datatype	MRI	PET	meeg	behavioral	microscopy	NIRS	motion
	anat	pet	eeg	beh	micr	nirs	motion
	func		meg				
	dwi		ieeg				
	perf						

```
ds001
├── dataset_description.json
├── participants.tsv
├── sub-01
│   ├── anat
│   │   ├── sub-01_inplaneT2.nii.gz
│   │   └── sub-01_T1w.nii.gz
│   └── func
│       ├── sub-01_task-balloonanalogrisktask_run-01_bold.nii.gz
│       ├── sub-01_task-balloonanalogrisktask_run-01_events.tsv
│       ├── sub-01_task-balloonanalogrisktask_run-02_bold.nii.gz
│       ├── sub-01_task-balloonanalogrisktask_run-02_events.tsv
│       ├── sub-01_task-balloonanalogrisktask_run-03_bold.nii.gz
│       └── sub-01_task-balloonanalogrisktask_run-03_events.tsv
├── sub-02
│   ├── anat
│   │   ├── sub-02_inplaneT2.nii.gz
│   │   └── sub-02_T1w.nii.gz
│   └── func
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│       ├── sub-02_task-balloonanalogrisktask_run-01_events.tsv
│       ├── sub-02_task-balloonanalogrisktask_run-02_bold.nii.gz
│       ├── sub-02_task-balloonanalogrisktask_run-02_events.tsv
│       ├── sub-02_task-balloonanalogrisktask_run-03_bold.nii.gz
│       └── sub-02_task-balloonanalogrisktask_run-03_events.tsv
├── ...
├── ...
└── task-balloonanalogrisktask_bold.json
```

code	participants.tsv	sub-MSC02	sub-MSC04	sub-MSC06	sub-MSC08	sub-MSC10	task-rest_bold.json
derivatives	sub-MSC01	sub-MSC03	sub-MSC05	sub-MSC07	sub-MSC09	task-motor_bold.json	

participants.tsv

participant_id	gender	age	education_degree	education_years
sub-MSC01	M	34	Doctorate	22
sub-MSC02	M	34	Doctorate	28
sub-MSC03	F	29	Masters	18
sub-MSC04	F	28	Bachelors	22
sub-MSC05	M	27	Bachelors	20
sub-MSC06	F	24	Bachelors	17.5
sub-MSC07	F	31	Masters	20
sub-MSC08	F	27	Professional	21
sub-MSC09	M	26	Professional	19
sub-MSC10	M	31	Professional	19

code	participants.tsv	sub-MSC02	sub-MSC04	sub-MSC06	sub-MSC08	sub-MSC10	task-rest_bold.json
derivatives	sub-MSC01	sub-MSC03	sub-MSC05	sub-MSC07	sub-MSC09	task-motor_bold.json	

task-motor_bold.json

```
"Manufacturer": "Siemens",
"ManufacturersModelName": "TrioTim",
"SoftwareVersion": "syngo MR B17",
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  0,
  1.1675,
  0.0625,
  1.2275,
  0.1225,
  1.29,
  :
]
```




SEARCH SUPPORT DOCUMENTATION [Sign in](#)

A free and open platform for validating and sharing BIDS-compliant [MRI](#), [PET](#), [MEG](#), [EEG](#), and [iEEG](#) data

63,347 Participants

1,475 Public Datasets

Browse by Modalities

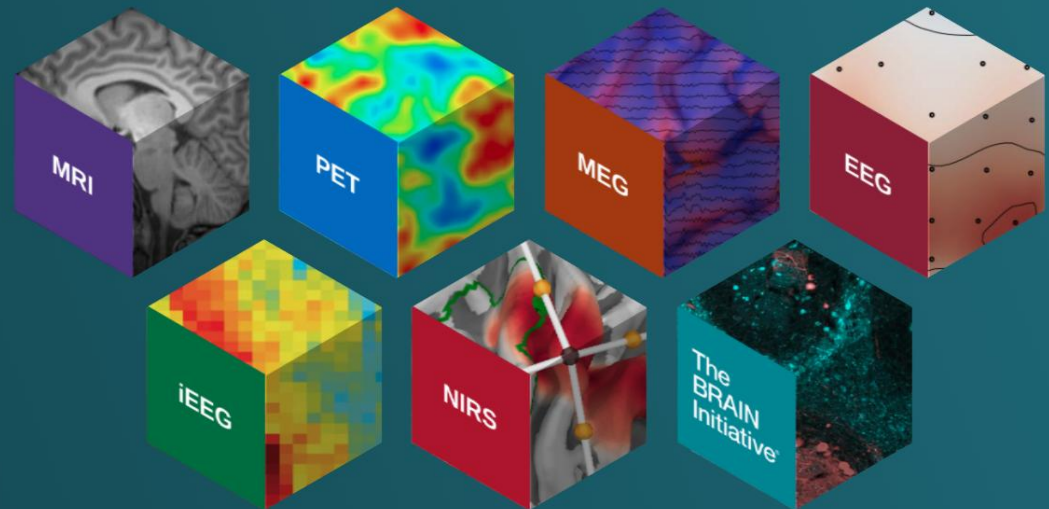


Or

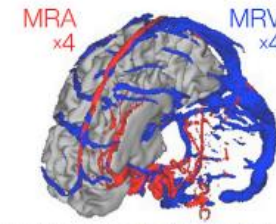
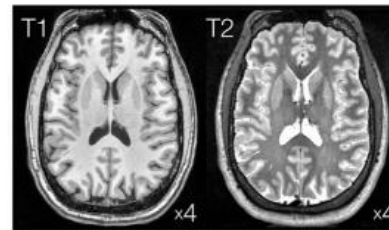
Search



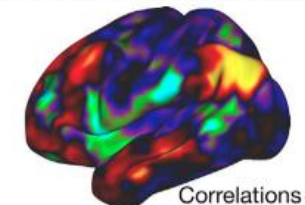
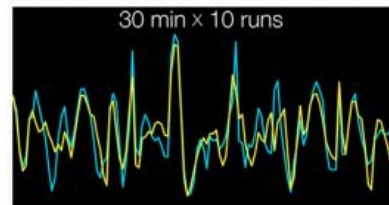
SIGN IN



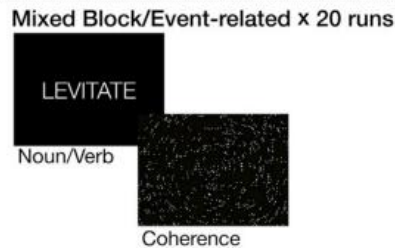
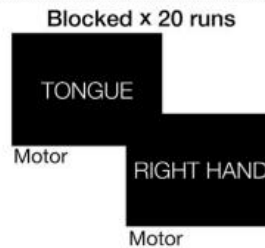
**Structural
MRI**



**RSFC
MRI**



**Task
fMRI**



**Neuropsych
Batteries**

Cognition



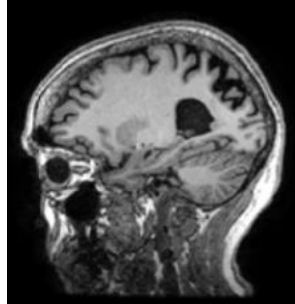
IQ



Personality

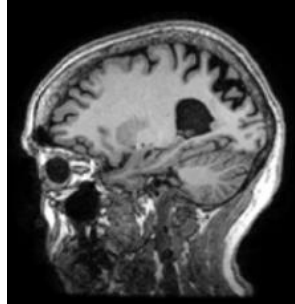


Anatomical
data

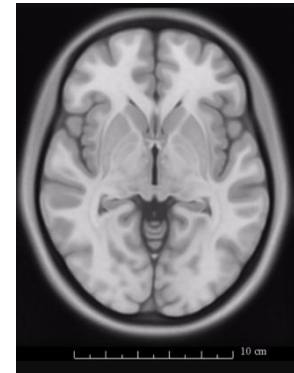


Many intertwined paths

Anatomical
data



normalization



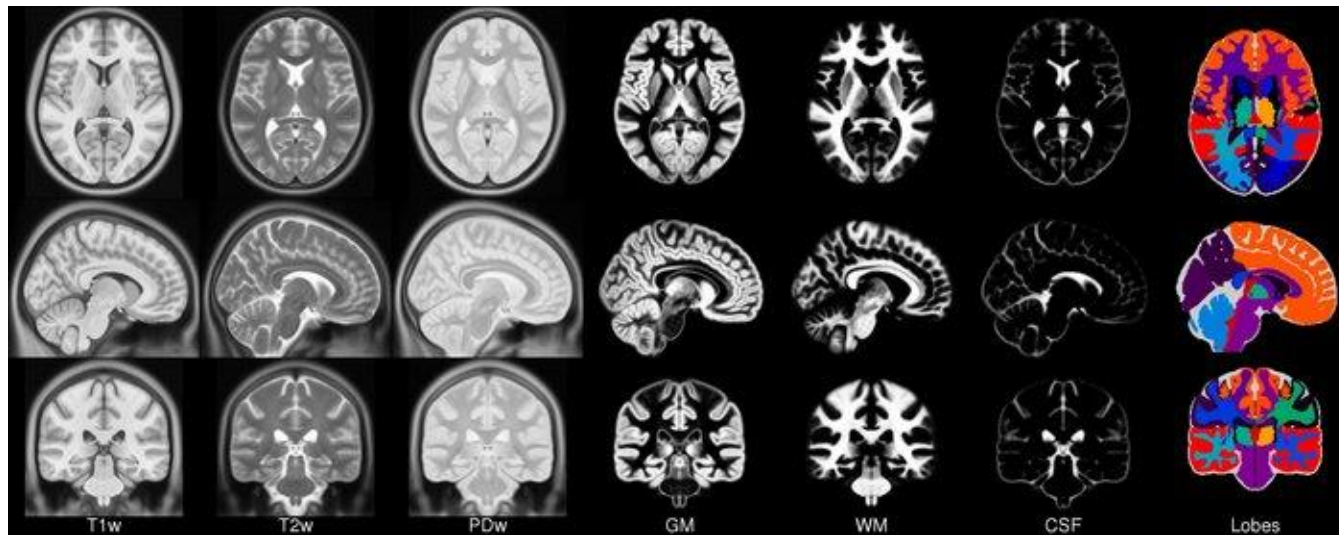
Anatomical
Template

Anatomical templates: MNI



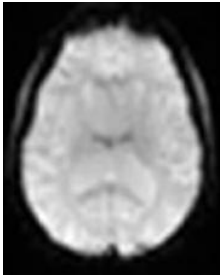
NeuroImaging
& Surgical Technologies Lab

<http://nist.mni.mcgill.ca/>



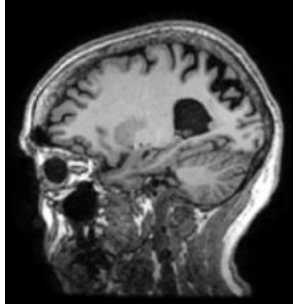
Many intertwined paths

Functional
data

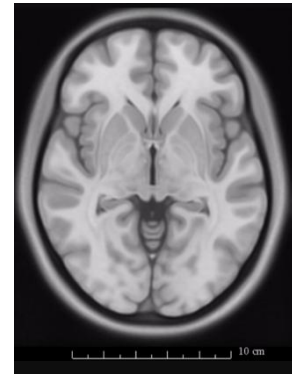


alignment

Anatomical
data

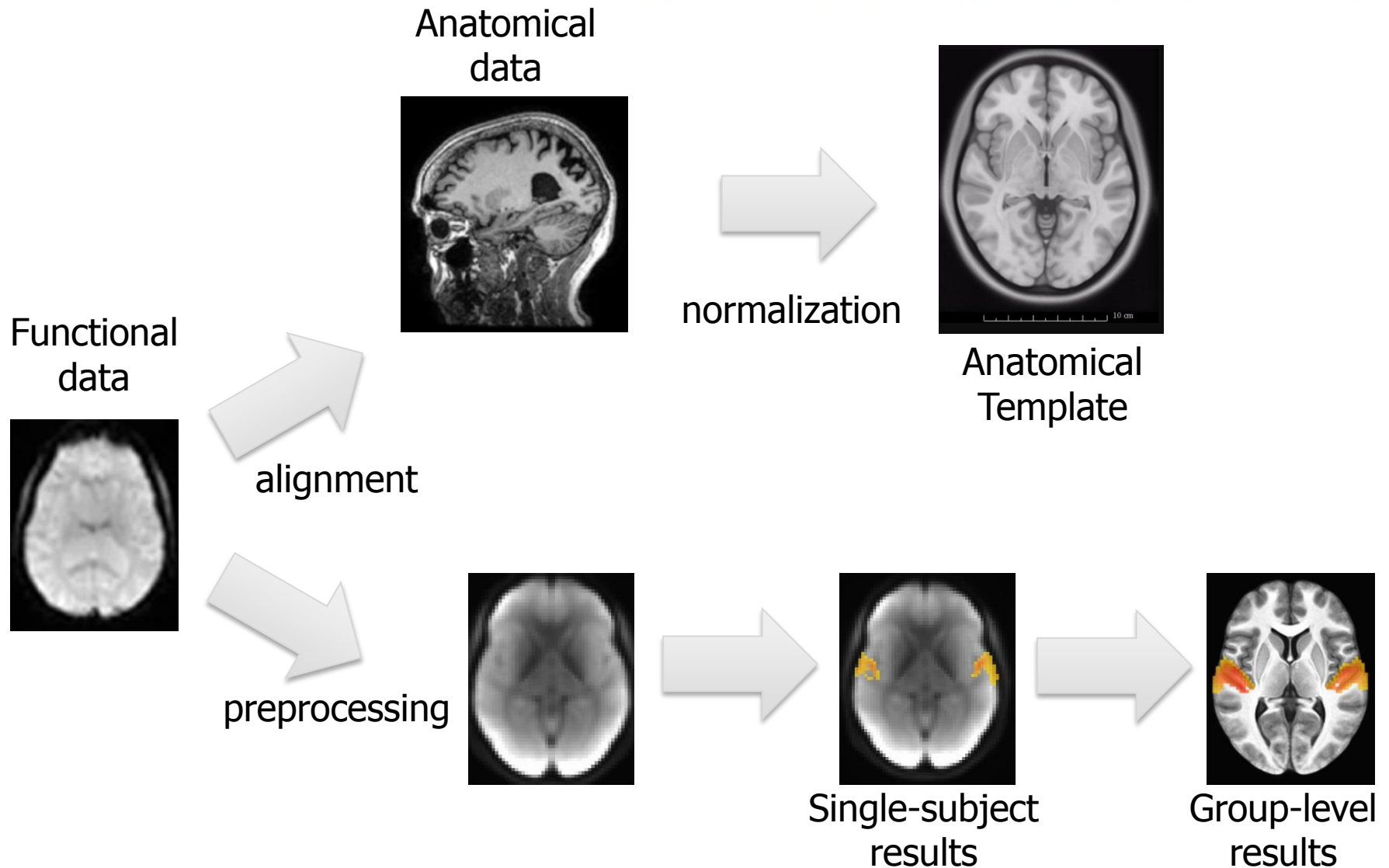


normalization



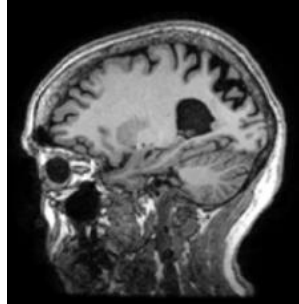
Anatomical
Template

Many intertwined paths

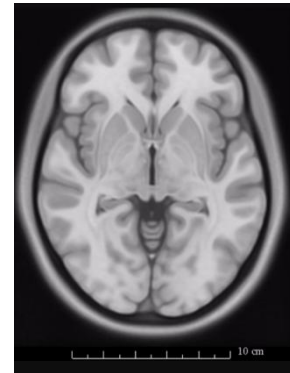


Many intertwined paths

Anatomical
data

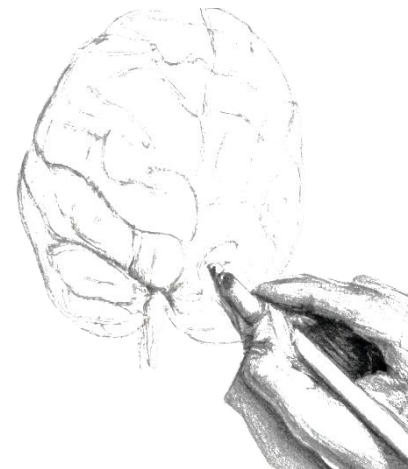


normalization



Anatomical
Template

code/processing_anat.sh

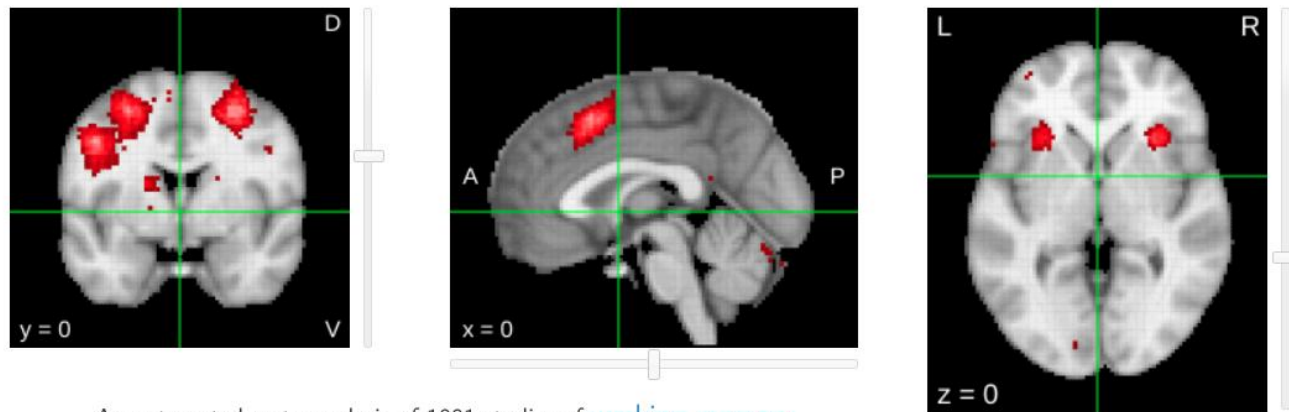


- **Cortical atlases** <https://www.lead-dbs.org/helpsupport/knowledge-base/atlasresources/cortical-atlas-parcellations-mni-space/>
- **Subcortical atlases** <https://www.lead-dbs.org/helpsupport/knowledge-base/atlasresources/atlas-2/>
- **The most useful Atlases (IMHO):**
 - Local-Global Parcellation of the Human Cerebral Cortex (Schaefer 2018)
 - HCP MMP 1.0 (Glasser 2016)
 - JuBrain / Juelich histological atlas (Eickhoff, Amunts, Zilles) <https://julich-brain-atlas.de/>
 - Harvard-Oxford cortical/subcortical atlases (Desikan 2006)

neurosynth.org

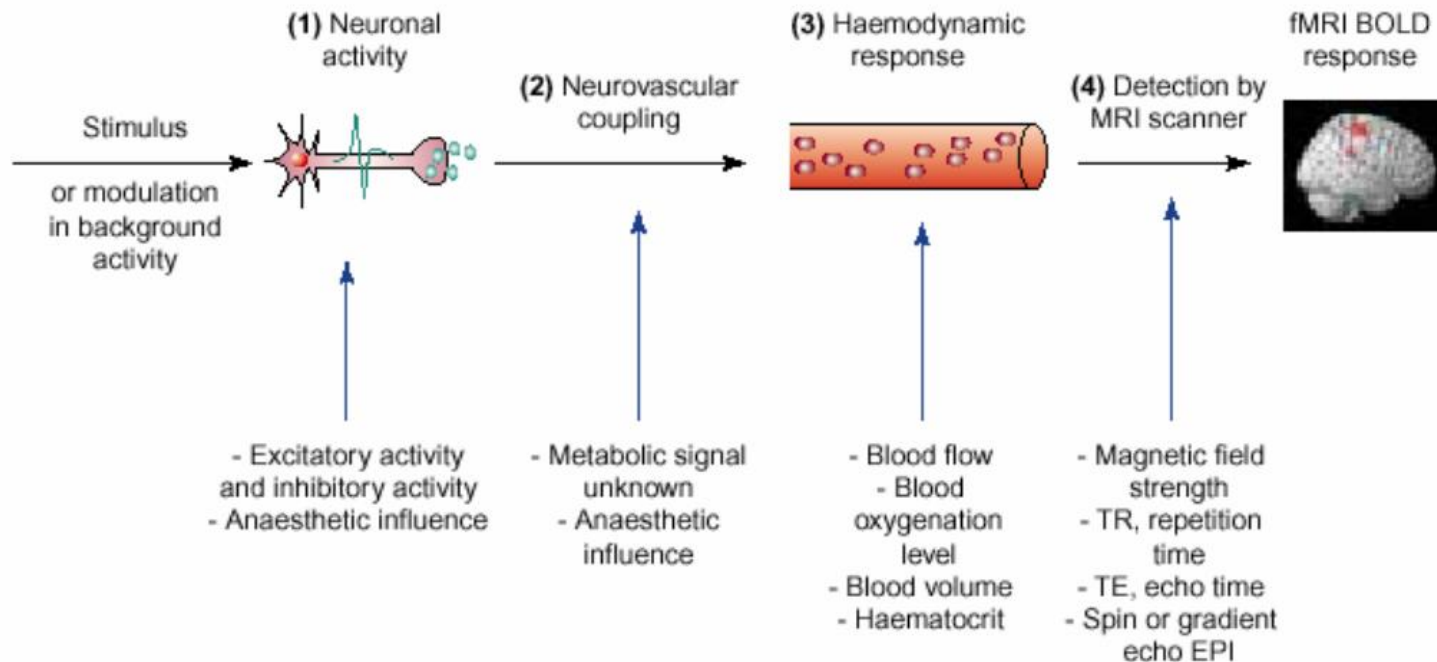
Neurosynth is a platform for large-scale, automated synthesis of functional magnetic resonance imaging (fMRI) data.

It takes thousands of published articles reporting the results of fMRI studies, chews on them for a bit, and then spits out images that look like this:

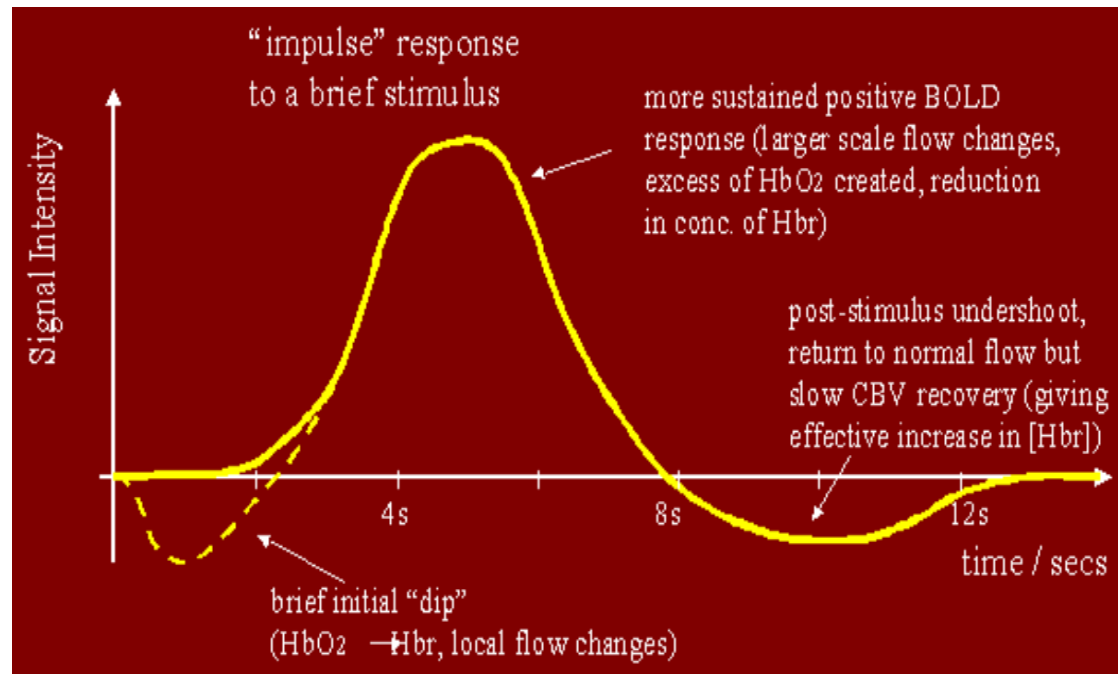


An automated meta-analysis of 1091 studies of [working memory](#)

BOLD: Blood Oxygenation Level Dependent

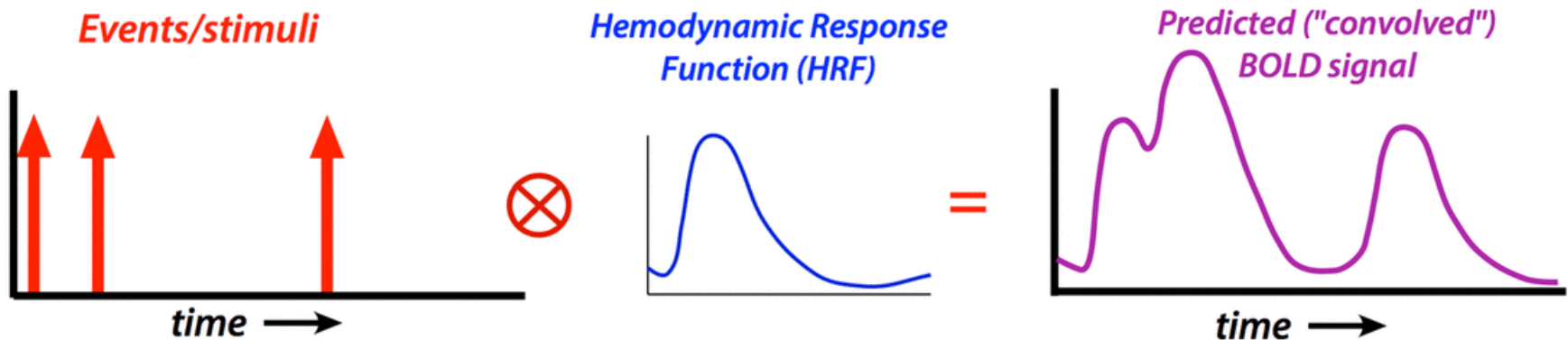


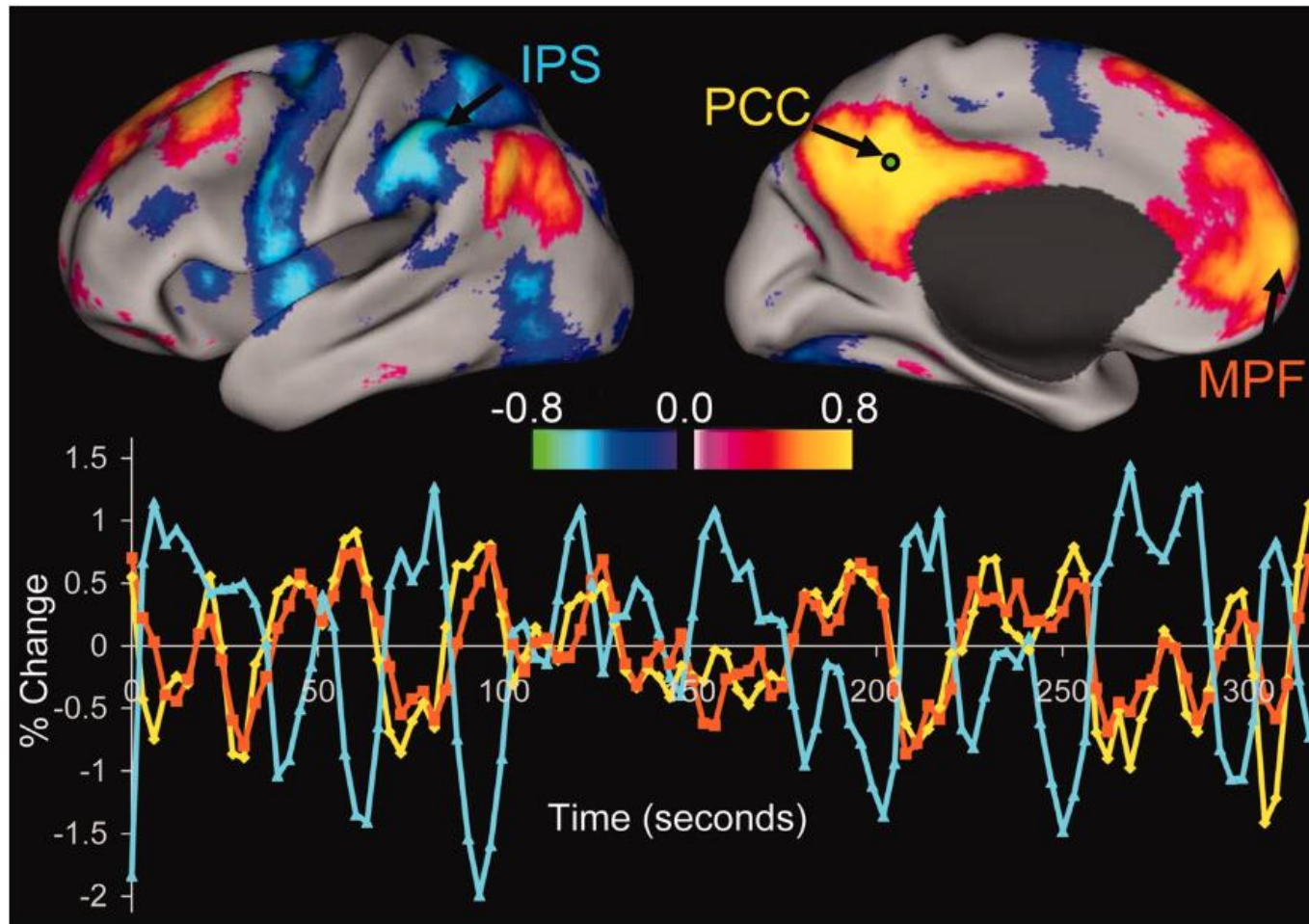
Hemodynamic Response Function (HRF)



HRF of multiple stimuli:

- accumulation of stimulus information over time (**summation**)
- response reduction with prolonged or repeated exposure (**adaptation**)





Respiration

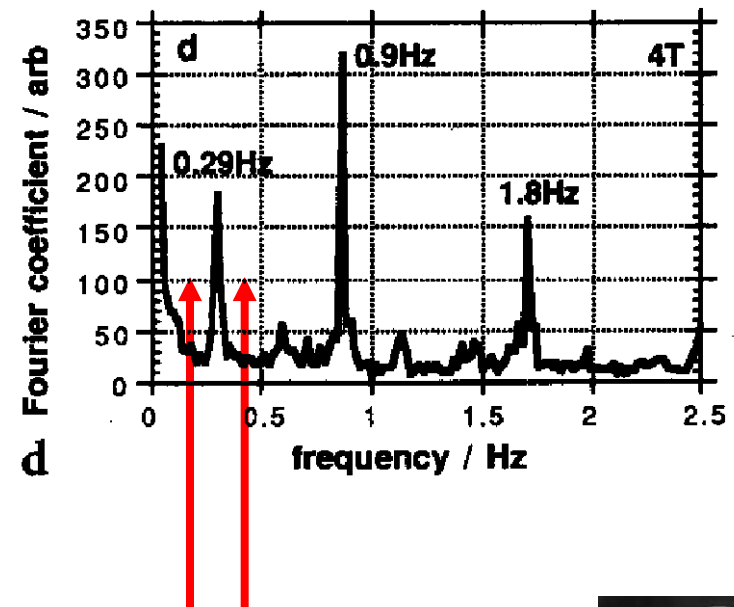
- every 4-10 sec (0.3 Hz)
- moving chest distorts susceptibility

Cardiac Cycle

- every ~1 sec (0.9 Hz)
- pulsing motion, blood changes

Solutions

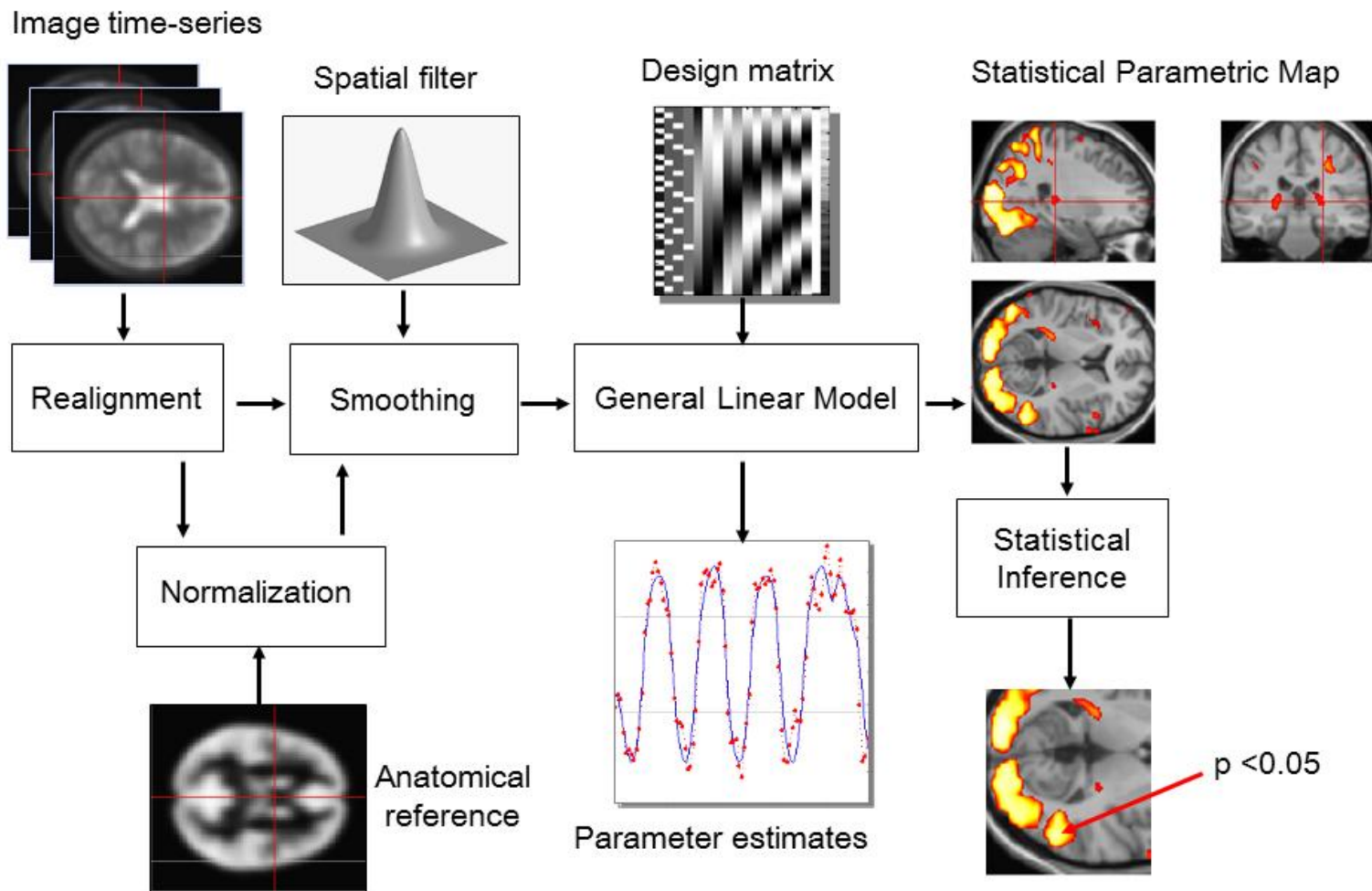
- gating
- avoiding paradigms at those frequencies

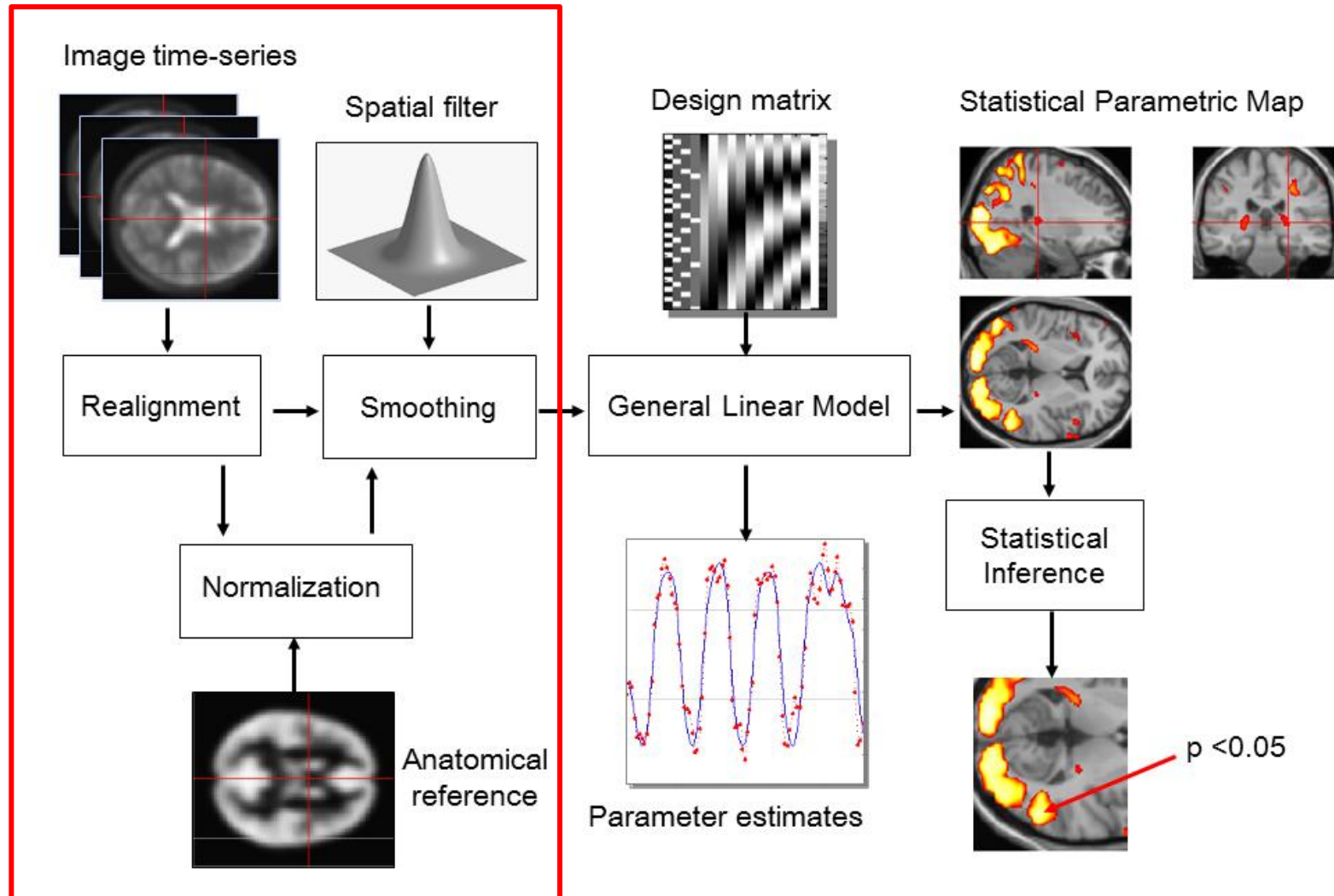


You want your paradigm frequency to be in a “sweet spot” away from the noise (pay attention to the Nyquist frequency!)



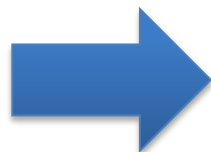
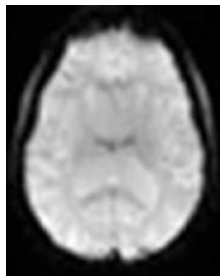
The (minimal) pipeline



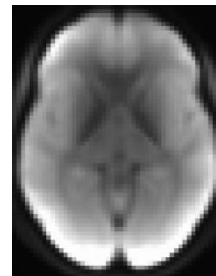


Functional preprocessing

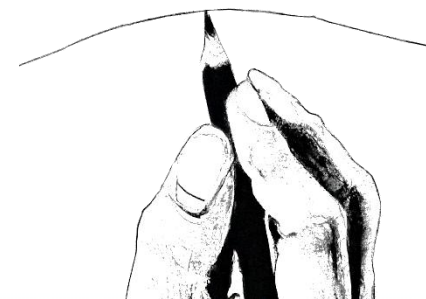
Functional data



preprocessing



code/processing_func.sh



Cognitive subtraction originated with reaction time experiments (F. C. Donders, a Dutch physiologist).

Measure the time for a process to occur by comparing two reaction times, one which has the same components as the other + the process of interest.

Example:

T1: Hit a button when you see a light

T2: Hit a button when the light is green but not red

T3: Hit the left button when the light is green and the right button when the light is red

$T2 - T1$ = time to make discrimination between light color

$T3 - T2$ = time to make a decision



Example in **fMRI**:

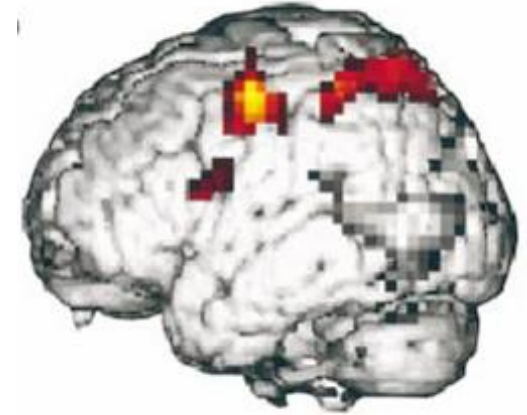
T1: View intact objects

T2: View scrambled objects

$T2 - T1$ = “object” areas

Possible interacting factors

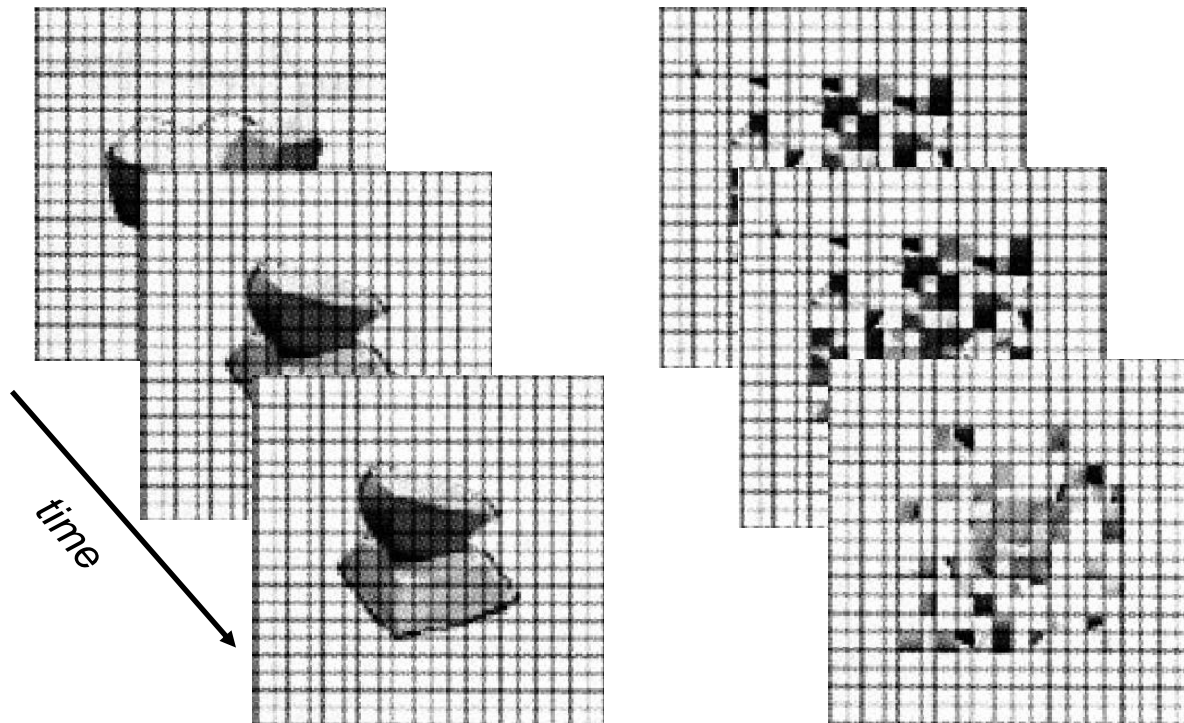
- *object-selective processing*
- *attentional salience* (objects are more interesting than scrambled objects)
- *memory* (you’ve seen an apple before but probably not a scrambled apple)



fMRI data seem highly susceptible to the amount of **attention** drawn to the stimulus or devoted to the task.

How can you ensure that activation is not simply due to an attentional confound?

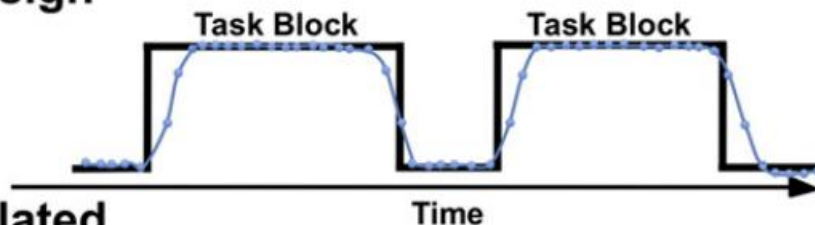
Add an attentional requirement to all stimuli or tasks!



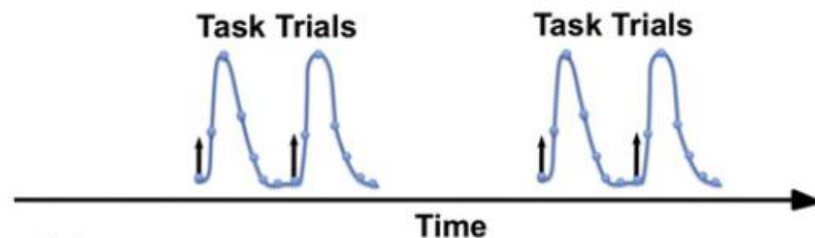
Add a “one back” task

- subject must hit a button whenever a stimulus repeats
- the repetition detection is much harder for the scrambled shapes
- any activation for the intact shapes cannot be due only to attention

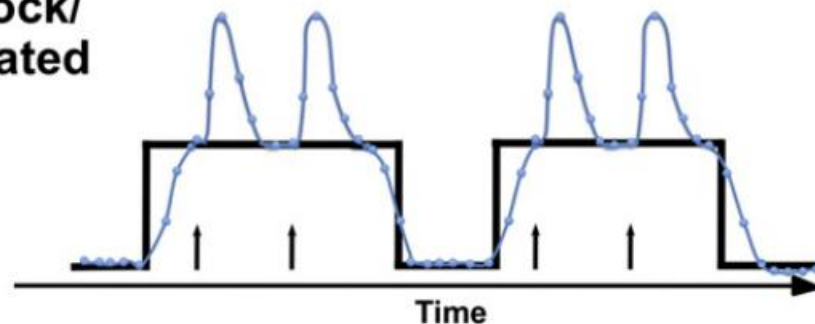
A) Block Design



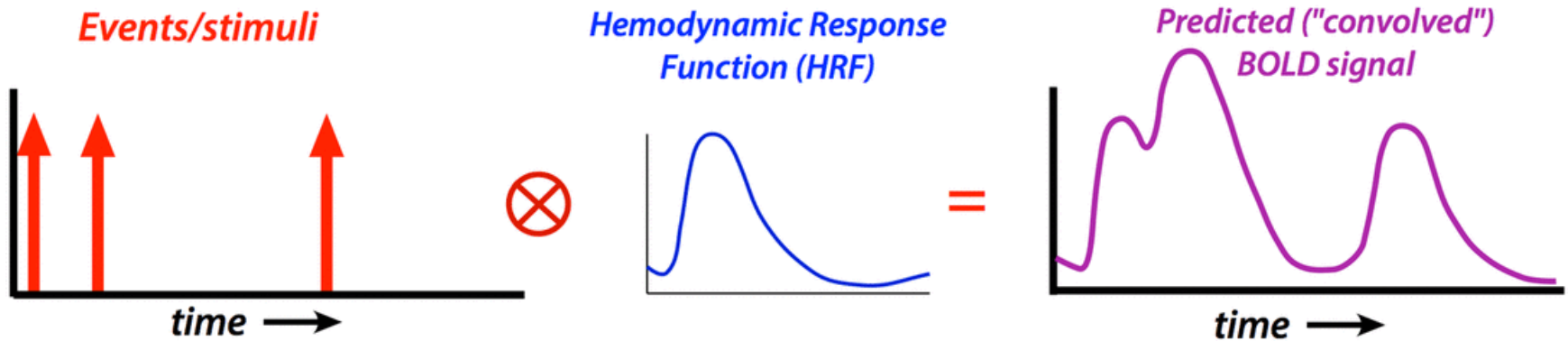
B) Event-Related



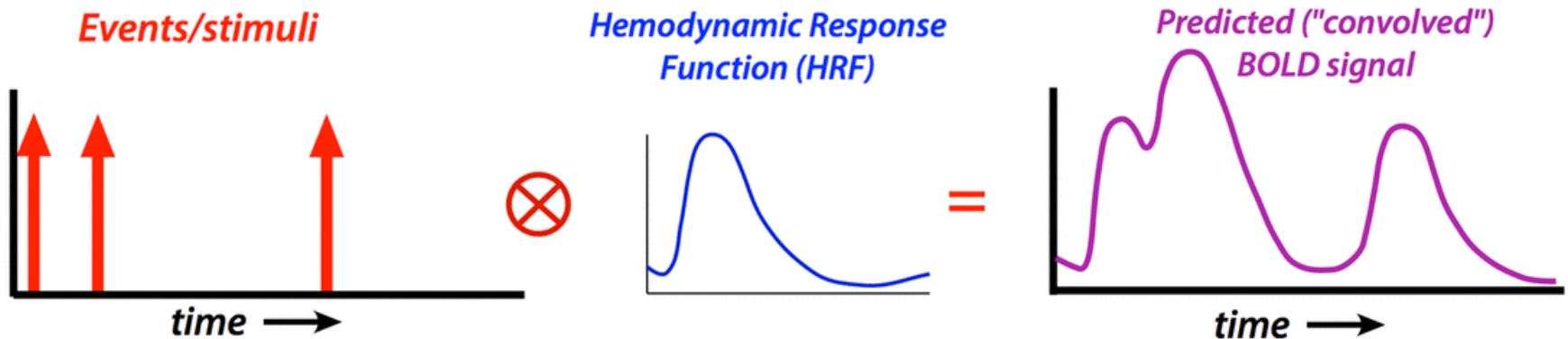
C) Mixed Block/Event-Related



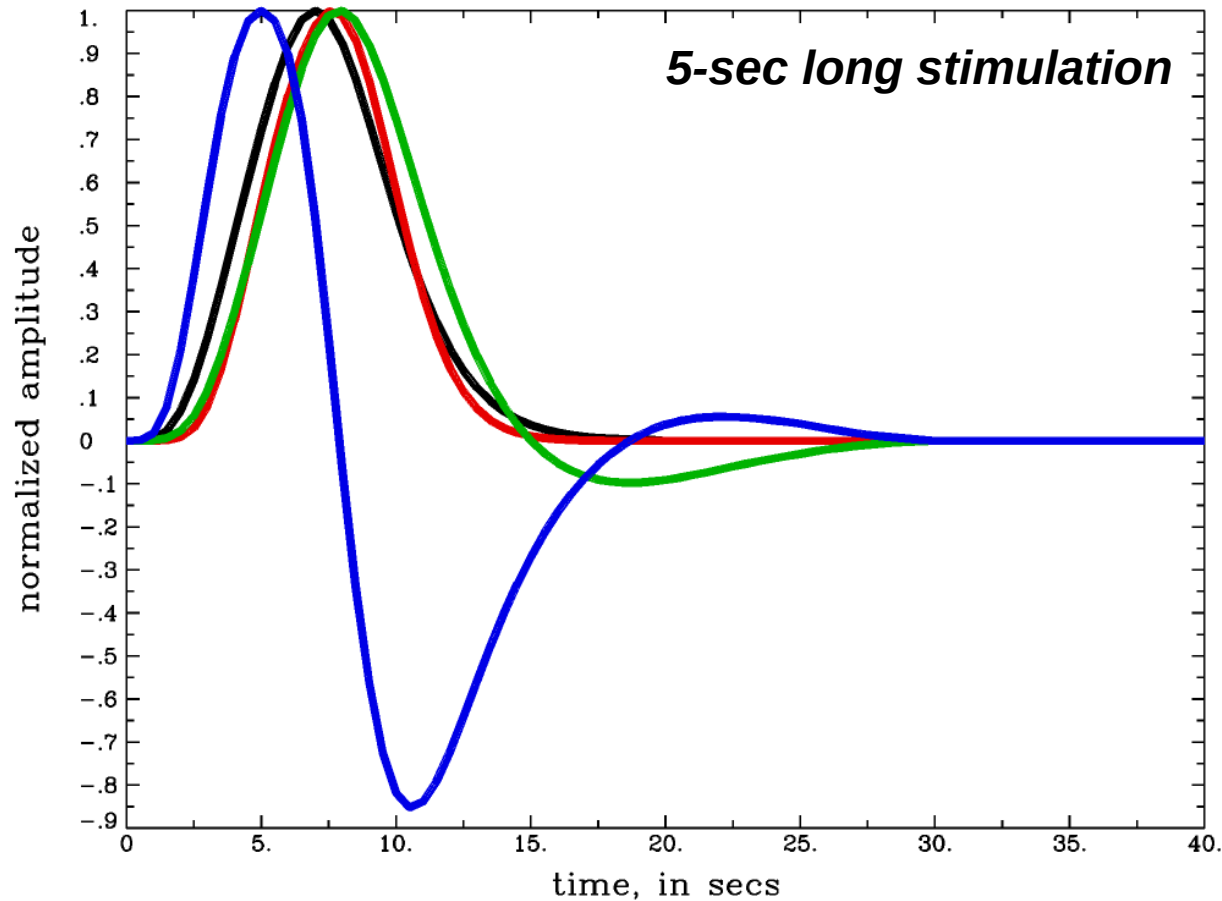
Temporal constraints of fMRI



- Boxcar functions for each trial (or condition) are convolved with HRF



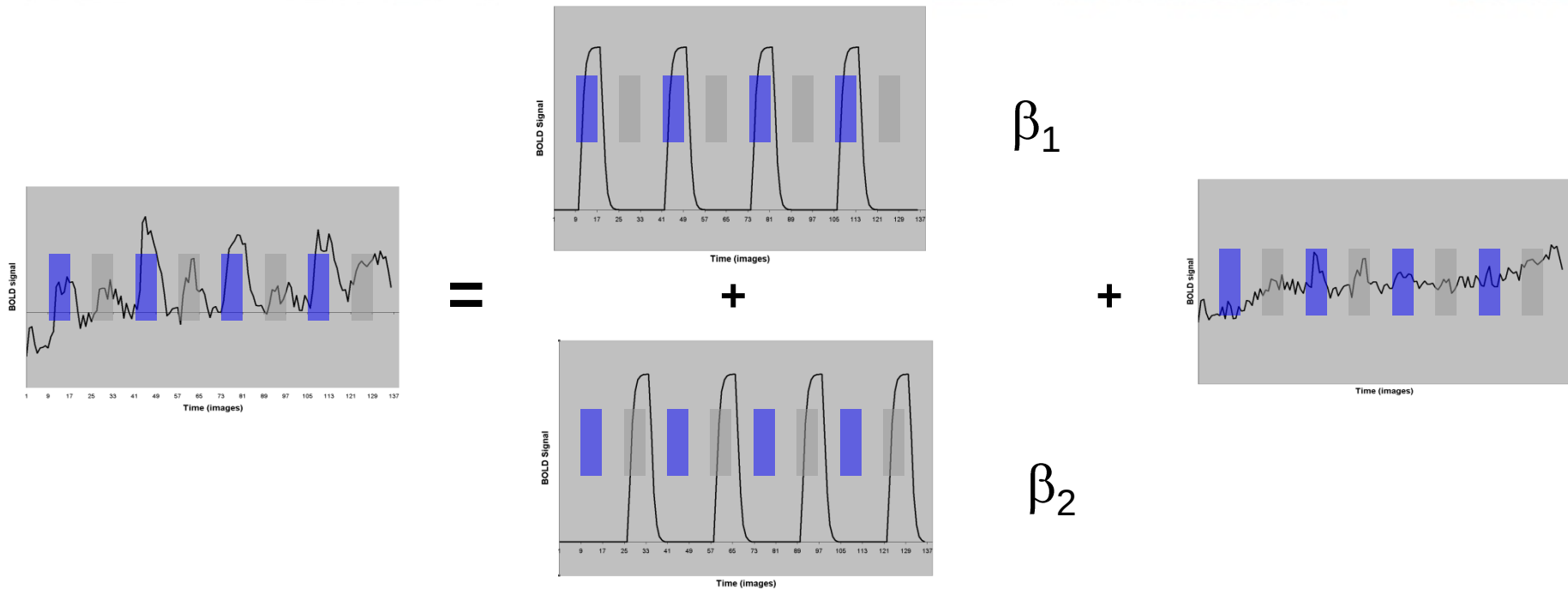
- The choice of the HRF is not trivial!



BLOCK

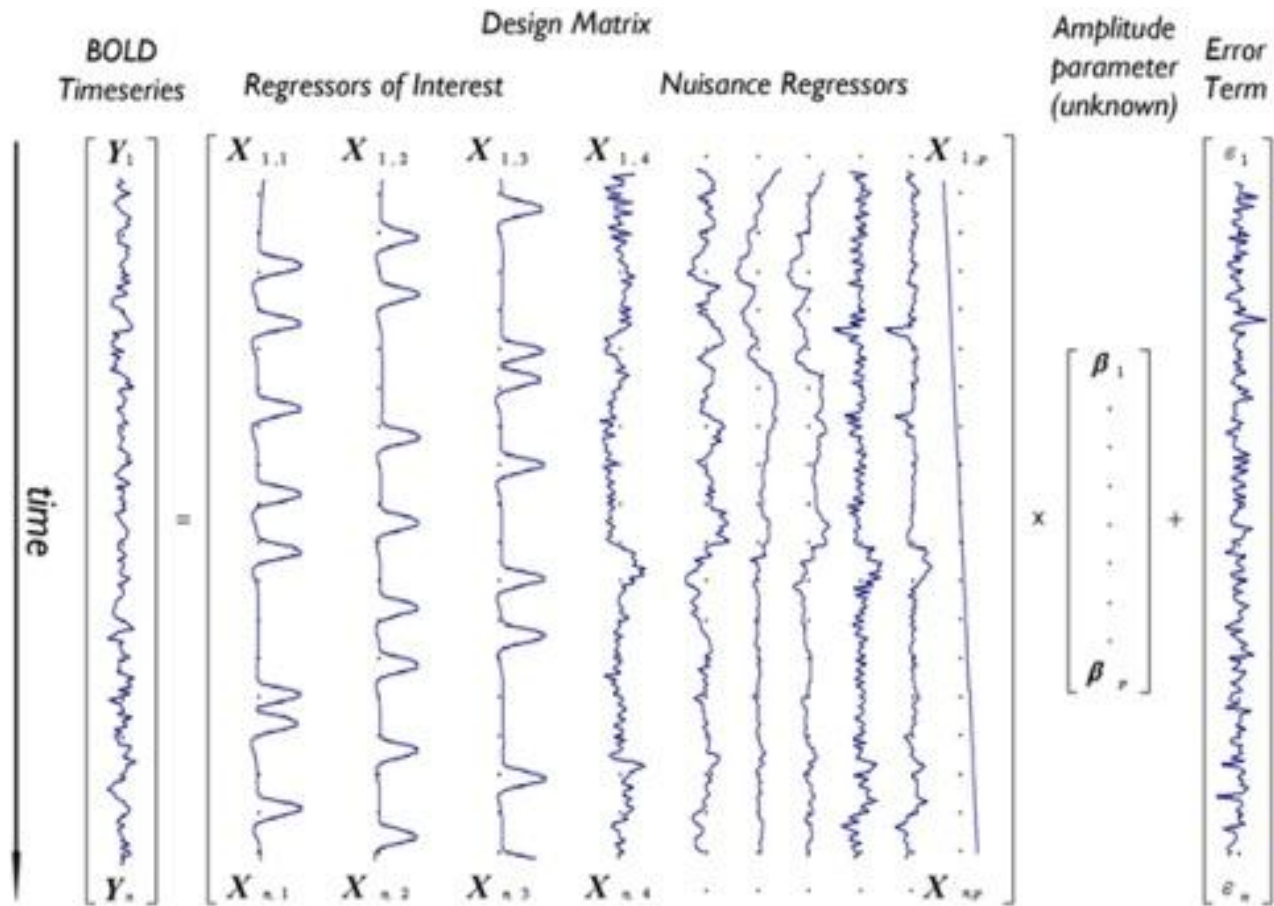
GAM

SPMG2 + TD



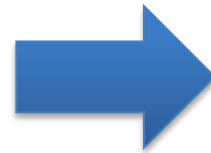
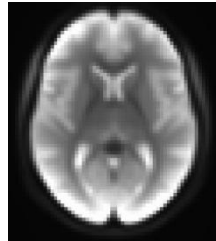
$$\begin{aligned}
 \text{fMRI Signal} &= \text{Design Matrix} \times \text{Betas} + \text{Residuals} \\
 \text{"our data"} &= \underbrace{\begin{matrix} \text{"what we CAN explain"} \\ \text{CAN} \end{matrix}} \times \underbrace{\begin{matrix} \text{"how much of it we CAN explain"} \\ \text{it we CAN explain"} \end{matrix}} + \underbrace{\begin{matrix} \text{"what we CANNOT explain"} \\ \text{CANNOT explain"} \end{matrix}}
 \end{aligned}$$

Statistical significance is basically a ratio of explained to unexplained variance

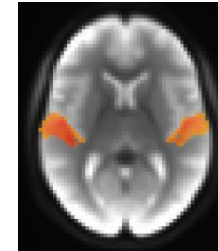


single-subject statistical analysis

Preprocessed
data



GLM



Results

[code/single_subject_analysis.sh](#)

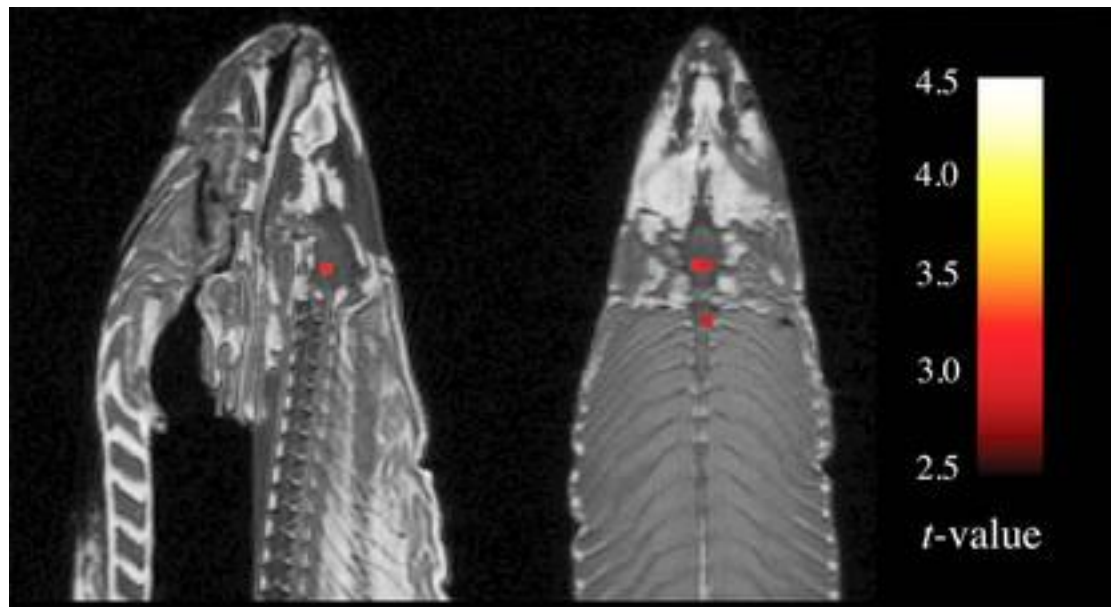


- GLM analysis returns:
 - GLT to contrast specific task conditions in the single participants
 - Beta coefficients (in % signal change) from regressors of interest that could be used as input from group-level analysis
 - Beta coeffs could be extracted for each individual trial (to be used in a machine-learning procedure)
 - T-stats that can be used in a Mixed Effects Meta Analysis
 - The residuals of a GLM without regressors of interest and by modeling ortvec only are considered the “cleaned” signals
 - GLM single-subject statistics are not robust!

Beta coefficients could be used in a voxel-wise statistical test:

- 3dttest++
- 3dMEMA (using also t-stats)
- 3dANOVA, 3dANOVA2, 3dANOVA3
- 3dRegAna
- FSL randomise

With the extreme dimensionality of functional neuroimaging data comes extreme risk for false positives. Across the 130,000 voxels in a typical fMRI volume the probability of a false positive with a family-wise significance threshold of 5% ($p < 0.05$) is certain

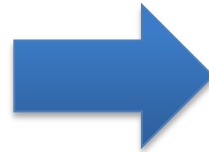
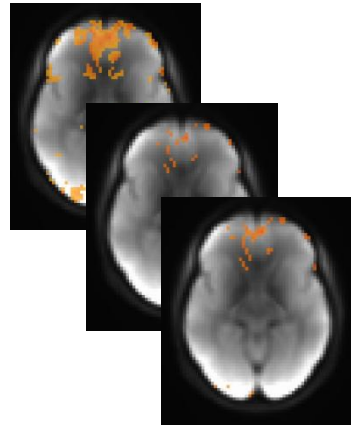


- The current way to count False Positive results is to analyze a large number of rest (null) datasets and simply count the number of datasets with significant activity. If a family-wise significance threshold of 5% is used, activity should be found in 5 out of 100 rest datasets. The null hypothesis does not state that there is not any brain activity in rest data (there is always activity in the brain), but that the rest data do not contain any brain activity that is correlated with a randomly selected regressor
- Currently, no method to count False Negatives is available!

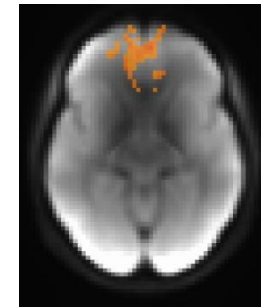
To control for False Positive results, there are several possibilities:

- Report results with a significant UNCORRECTED p-value ($p < 0.001$)
- Cluster-based corrections (e.g., AFNI's ClustSim, FSL's TFCE, Smith et al., 2009)
- Correct for False Discovery Rate (FDR, Benjamini & Hochberg, 1995)
- Family-wise error correction (FWE, Bonferroni correction or permutation-based correction -Nichols et al., 2002-)

Single Subjects



Statistical Test

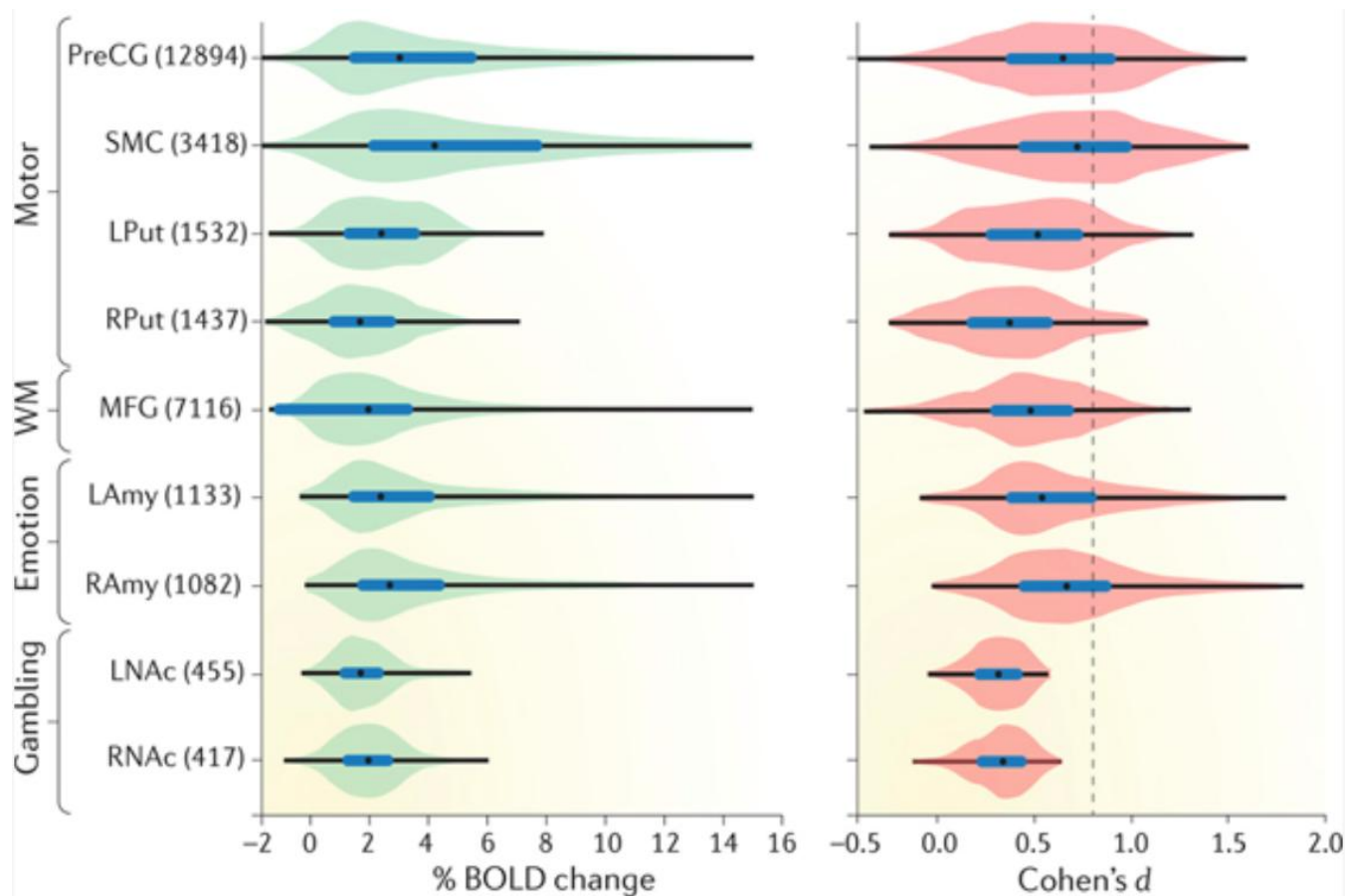


Group Level
Results

`code/group_analysis.sh`



fMRI: small effect size



- Analytical flexibility can have substantial effects on scientific conclusions. We emphasize the importance of validating and sharing complex analysis workflows, and the need for performing and reporting multiple analyses of the same data

Article

Variability in the analysis of a single neuroimaging dataset by many teams

<https://doi.org/10.1038/s41586-020-2314-9>

Received: 14 November 2019

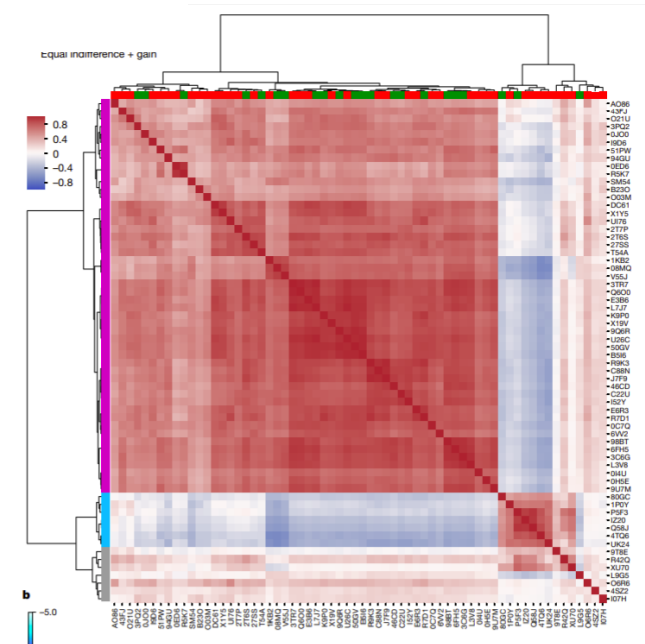
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Published online: 20 May 2020

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A list of authors and affiliations appears in the online version of the paper.

Data analysis workflows in many scientific domains have become increasingly complex and flexible. Here we assess the effect of this flexibility on the results of functional magnetic resonance imaging by asking 70 independent teams to analyse the same dataset, testing the same 9 ex-ante hypotheses¹. The flexibility of analytical approaches is exemplified by the fact that no two teams chose identical workflows to analyse the data. This flexibility resulted in sizeable variation in the results of hypothesis tests, even for teams whose statistical maps were highly correlated at intermediate stages of the analysis pipeline. Variation in reported results was related to several aspects of analysis methodology. Notably, a meta-analytical approach that aggregated information across teams yielded a significant consensus in activated regions. Furthermore, prediction markets of researchers in the field revealed an overestimation of the likelihood of significant findings, even by researchers with direct knowledge of the dataset^{2–5}. Our findings show that analytical flexibility can have substantial effects on scientific conclusions, and identify factors that may be related to variability in the analysis of functional magnetic resonance imaging. The results emphasize the importance of validating and sharing complex analysis workflows, and demonstrate the need for performing and reporting multiple analyses of the same data. Potential approaches that could be used to mitigate issues related to analytical variability are discussed.



- Brain-behavior associations were smaller than thought, resulting in statistically underpowered studies, inflated effect sizes and replication failures at typical sample sizes. As sample sizes grew into the thousands, replication rates began to improve and effect size inflation decreased

Article

Reproducible brain-wide association studies require thousands of individuals

<https://doi.org/10.1038/s41586-022-04492-9>

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Check for updates

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Magnetic resonance imaging (MRI) has transformed our understanding of the human brain through well-replicated mapping of abilities to specific structures (for example, lesion studies) and functions^{1–3} (for example, task functional MRI (fMRI)). Mental health research and care have yet to realize similar advances from MRI. A primary

